AN EQUIVALENCE OF MODULE CATEGORIES IS TENSORING WITH AN INVERTIBLE MODULE OVER A RING ISOMORPHISM A → B

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file functors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for functors.tex (fu-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0GNS: sources/stacks-chow/tags.txt, source lines 20802–20802. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0GNW: sources/stacks-chow/tags.txt, source lines 20806–20806. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GNV: sources/stacks-chow/tags.txt, source lines 20805–20805. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B8I: sources/stacks-chow/tags.txt, source lines 14141–14141. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NX: sources/stacks-chow/tags.txt, source lines 875–875. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let A and B be rings. If*

$$F : Mod_A \longrightarrow Mod_B$$

is an equivalence of categories, then there exists an isomorphism $A \to B$ of rings and an invertible B-module L such that F is isomorphic to the functor $M \mapsto (M \otimes_A B) \otimes_B L$.

Proof. Since an equivalence commutes with all colimits, we see that Lemmas 0GNS applies. Let K be the $A \otimes_{\mathbf{Z}} B$-module such that F is isomorphic to the functor $M \mapsto M \otimes_A K$. Let K' be the $B \otimes_{\mathbf{Z}} A$-module such that a quasi-inverse of F is

Date: stacks-project:functors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

isomorphic to the functor $N \mapsto N \otimes_B K'$. By Remark 0GNW and Lemma 0GNV we have an isomorphism

$$\psi : K \otimes_B K' \longrightarrow A$$

of $A \otimes_{\mathbf{Z}} A$-modules. Similarly, we have an isomorphism

$$\psi' : K' \otimes_A K \longrightarrow B$$

of $B \otimes_{\mathbf{Z}} B$-modules. Choose an element $\xi = \sum_{i=1,\ldots,n} x_i \otimes y_i \in K \otimes_B K'$ such that $\psi(\xi) = 1$. Consider the isomorphisms

$$K \xrightarrow{\psi^{-1} \otimes \mathrm{id}_K} K \otimes_B K' \otimes_A K \xrightarrow{\mathrm{id}_K \otimes \psi'} K$$

The composition is an isomorphism and given by

$$k \longmapsto \sum x_i \psi'(y_i \otimes k)$$

We conclude this automorphism factors as

$$K \to B^{\oplus n} \to K$$

as a map of B-modules. It follows that K is finite projective as a B-module.

We claim that K is invertible as a B-module. This is equivalent to asking the rank of K as a B-module to have the constant value 1, see More on Algebra, Lemma 0B8I and Algebra, Lemma 00NX. If not, then there exists a maximal ideal $\mathfrak{m} \subset B$ such that either (a) $K \otimes_B B/\mathfrak{m} = 0$ or (b) there is a surjection $K \to (B/\mathfrak{m})^{\oplus 2}$ of B-modules. Case (a) is absurd as $K' \otimes_A K \otimes_B N = N$ for all B-modules N. Case (b) would imply we get a surjection

$$A = K \otimes_B K' \longrightarrow (B/\mathfrak{m} \otimes_B K')^{\oplus 2}$$

of (right) A-modules. This is impossible as the target is an A-module which needs at least two generators: $B/\mathfrak{m} \otimes_B K'$ is nonzero as the image of the nonzero module $B/\mathfrak{m}$ under the quasi-inverse of F.

Since K is invertible as a B-module we see that $\mathrm{Hom}_B(K, K) = B$. Since $K = F(A)$ the action of A on K defines a ring isomorphism $A \to B$. The lemma follows. $\square$